

Analysis of the Obstacle Potential Function

1 Potential Function

The obstacle potential is defined as

$$U_{obst} = \frac{\lambda(\Delta v / \Delta v_{max})}{\left(1 - \frac{w}{\Delta x}\right) \sqrt{\left(\frac{\Delta x}{\tau(\Delta x, \Delta v_x)}\right)^2 + \left(\frac{\Delta y}{\tau(\Delta y, \Delta v_y)}\right)^2}} \quad (1)$$

where the direction-dependent time constants are given by

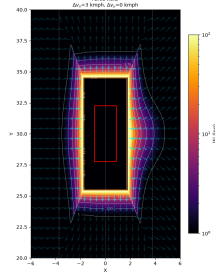
$$\tau(\Delta x, \Delta v_x) = \frac{(1 + \alpha \Delta v_x) + (\alpha \Delta v_x - 1) \tanh(-\Delta x \Delta v_x)}{2} \tau_x \quad (2)$$

$$\tau(\Delta y, \Delta v_y) = \frac{(1 + \alpha \Delta v_y) + (\alpha \Delta v_y - 1) \tanh(-\Delta y \Delta v_y)}{2} \tau_y \quad (3)$$

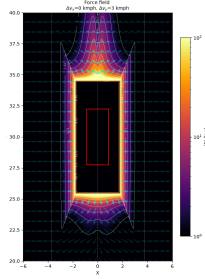
Here $\Delta x, \Delta y$ are the relative position offsets to the obstacle vehicle, $\Delta v_x, \Delta v_y$ the relative velocities, w the obstacle width, and $\lambda, \tau_x, \tau_y, \alpha, \Delta v_{max}$ are model parameters.

1.1 Force Field

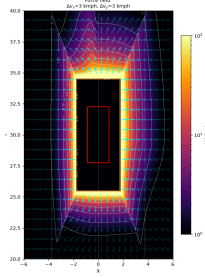
The force field $\mathbf{F} = -\nabla U_{obst}$ resulting from Equation 1 is shown below, decomposed into its x - and y -components and as the combined field.



(a) v_x



(b) v_y



(c) Both

Figure 1: Force field of the potential in Equation 1.

2 Force at Fixed Points vs. Relative Velocity

To examine the velocity dependence of the force in more detail, we fix the position and sweep the relative velocity, plotting the resulting force magnitude for several directions of $(\Delta v_x, \Delta v_y)$. We consider the point $(x, y) = (3, 30)$, i.e. ahead of and to the side of the obstacle, and the point $(x, y) = (-3, 30)$, on the opposite side.

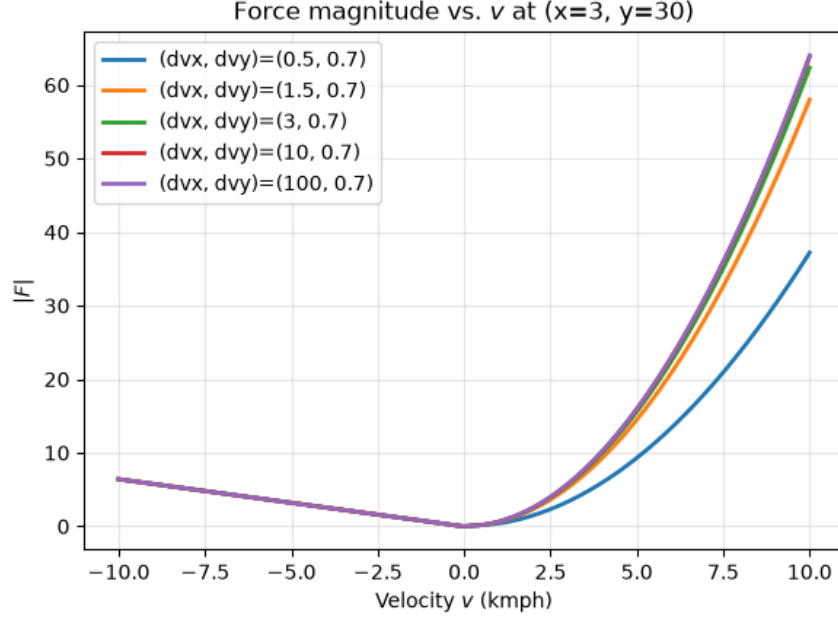


Figure 2: Force magnitude vs. relative velocity at $(x, y) = (3, 30)$ and $(x, y) = (-3, 30)$.

3 Removing the $\Delta v / \Delta v_{max}$ Term

From Figure 2, we observe that when the vehicle is receding, an increase in the relative velocity causes the force to increase as well. This is an undesirable effect, since a receding vehicle should not produce a growing repulsive force.

To avoid this behaviour, we drop the $\Delta v / \Delta v_{max}$ term from the original potential function in Equation 1. Repeating the same two experiments with this modified potential gives the plot below.

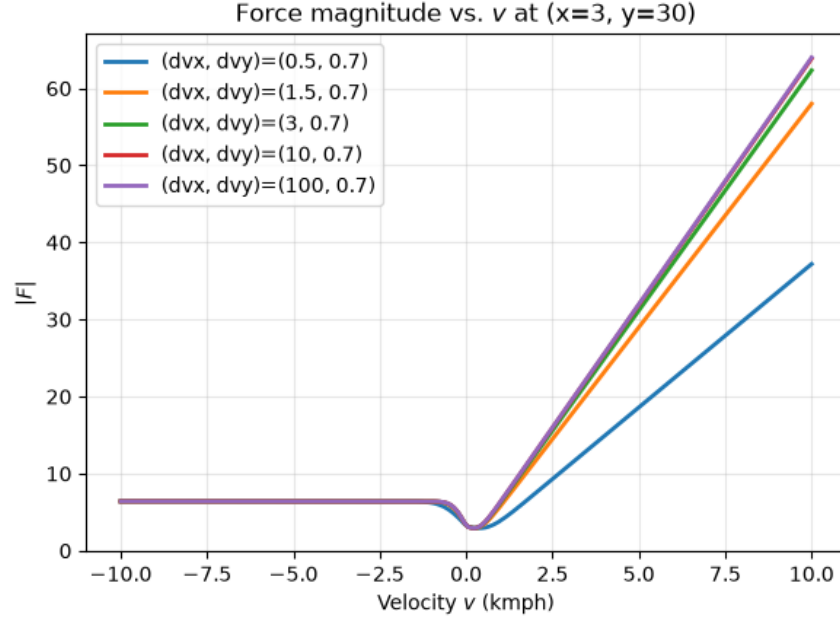


Figure 3: Force magnitude vs. relative velocity at $(x, y) = (3, 30)$ and $(x, y) = (-3, 30)$, with the $\Delta v / \Delta v_{max}$ term removed.